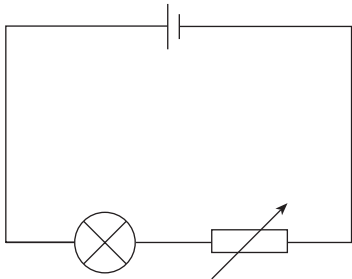


3. (a) Students are investigating the current-voltage characteristic of a filament lamp. An incomplete diagram of their circuit is shown below.



- (i) **Add to the diagram** a correctly connected ammeter and voltmeter. [2]
- (ii) Describe how the circuit is used to obtain a series of measurements of current and voltage. [3]

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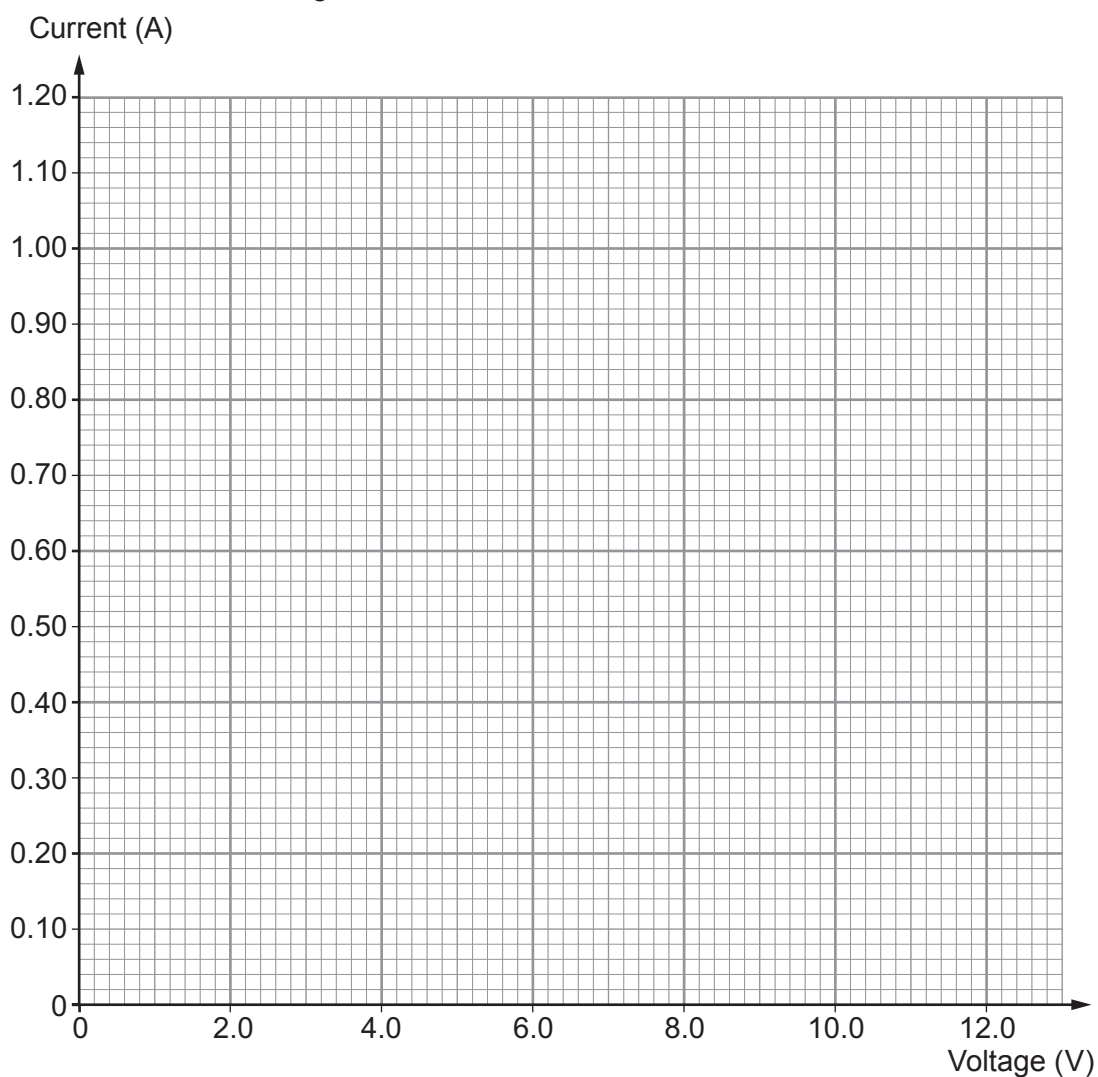
- (b) The following data were collected using the circuit.

Voltage (V)	Current (A)
0.0	0.00
2.0	0.46
4.0	0.76
6.0	0.94
8.0	1.02
10.0	1.06
12.0	1.08



Plot the data on the grid below and draw a suitable line.

[3]



(c) (i) Use the graph and the equation:

$$\text{resistance} = \frac{\text{voltage}}{\text{current}}$$

to calculate the resistance of the lamp when the voltage across it is 3 V.

[2]

Resistance = Ω

(ii) The students conclude that the **resistance** of the lamp increases as voltage increases. Explain how the data support this conclusion.

[2]

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